

Bach Nguyen

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EDUCATION

Ontario Tech University

BSc (Hons), Integrated Mathematics & Computer Science (Co-op)

Oshawa, ON

Sep. 2023 – Apr. 2027 (Expected)

- Cumulative GPA: 3.77/4.3 | President's List (3x), Dean's List
- Relevant Coursework: Data Mining, Database Systems, Scientific Data Analysis, Data Structures, Statistics & Probability

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, Bash

Data & Development: ETL Pipelines, Microsoft Excel, Power Query, pandas, NumPy, scikit-learn, SciPy, Matplotlib

Databases: MySQL, SQLite, IBM DB2

Tools & Platforms: Azure, SharePoint, Power Apps, Power Automate, Visual Studio, Docker, Apache Airflow, Git, Jupyter, BeautifulSoup

Certifications: Microsoft Office Specialist (Word); IBM Z Xplore – Concept Batch (z/OS, JCL, COBOL); IBM Python for Data Engineering

PROFESSIONAL EXPERIENCE

IT Intern

May 2025 – Aug. 2025

Canadian Mental Health Association

Oshawa, ON

- Managed employee data and company files in SharePoint, and consolidated staff records into Excel workbooks with lookup formulas and pivot tables for direct manager reporting.
- Built and deployed automated booking workflows with Power Apps and Power Automate, eliminating recurring double-booking errors and saving staff an estimated 5+ hours of manual scheduling per week.
- Validated workflow outputs and data accuracy through structured testing before deployment.
- Modernized legacy internal tools, gathering requirements from non-technical staff and documenting the handoff for ongoing IT support.
- Maintained servers and resolved tier-1/tier-2 IT issues for 100+ staff, sustaining uptime for clinical software critical to client services.

Undergraduate Thesis Researcher, Visual Computing Lab

Sep. 2026 – Present

Ontario Tech University — supervised by Dr. Faisal Qureshi

Oshawa, ON

- Exploring continual learning methods for parameter-efficient vision transformers, building on the lab's CascadedViT architecture.
- Implemented a 10M-parameter CascadedViT variant under CPG (Compacting, Picking, and Growing); collaborating with a CascadedViT co-author on the expert-routing mechanism in SEMA.

Teaching Assistant, Prep. Math for Engineers (MATH-0900U)

Sep. 2026 – Present

Ontario Tech University

Oshawa, ON

- Lead weekly tutorials for a first-year engineering mathematics course, invigilate exams, and grade all tutorial work, midterms, and finals.

Data Engineer (Volunteer)

Feb. 2024

Statistics Without Borders — Project #279 (IPÊ, Institute for Ecological Research)

Remote

- Designed Azure data pipelines to organize large-scale biodiversity datasets for a cross-disciplinary team of statisticians and scientists.
- Audited cloud storage practices and designed an optimized data-organization strategy to reduce query times and storage costs.

PROJECTS

Hospital Readmission Risk Model | Python, XGBoost, LightGBM, SHAP

2025

- Modeled 30-day readmission risk across ~100k patient encounter records, deduplicating repeat patients and encoding high-cardinality diagnosis and medication fields.
- Used chronological train/test splits and AUPRC-first scoring on a 9% positive class, achieving 0.128 AUPRC under leakage-free validation.
- Applied SHAP to identify leading risk drivers, including admission complexity, prior admissions, and medication count.

Global GDP ETL Pipeline | Python, BeautifulSoup, SQLite

2025

- Built an end-to-end ETL pipeline to scrape, clean, and normalize international economic data into a structured SQLite database.
- Designed a relational schema consolidating inconsistent multi-source records into a single queryable table for longitudinal analysis.

LEADERSHIP & COMMUNITY INVOLVEMENT

Events Coordinator, Ontario Tech Computer Science Club

Sep. 2025 – Present

- Co-organized an annual hackathon with 250+ participants, leading logistics, sponsor relations, and event-day operations.